AN EXTENSION OF THE DUALIZING SHEAF BY A ZERO-DIMENSIONAL QUOTIENT WITH AT LEAST TWO SECTIONS IS GLOBALLY GENERATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the rank-1 setup, the H^1 duality step, the self-map closure argument, the author's marked automorphism fact used as a step, and the final contradiction are all retained verbatim.

Editorial note. The label lemma-globally-generated-curve names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The globally generated subsheaf lemma. Same chapter, Lemma lemma-globally-generated. Shows the subsheaf F' generated by global sections already surjects onto Q, so F' is nonzero.
- Duality on proper curves. Duality for Schemes, Lemma duality-lemma-dualizing-module-proper-over-A. Turns nonzero H^1(X, F') into a nonzero map F' -> omega_X in the contradiction step.

Source reference. Original lemma 0AWP: sources/stacks-chow/tags.txt, source lines 13727–13727. Prerequisite pointer invoked at proof line(s) 1426; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 01AN: sources/stacks-curves/pinned-original-source.tex, source lines 1372–1372. Prerequisite pointer invoked at proof line(s) 1418; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0B5D: sources/stacks-curves/pinned-original-source.tex, source lines 1307–1307. Prerequisite pointer invoked at proof line(s) 1408; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 0B5D assume that X is integral. Let $0 \to \omega_X \to \mathcal{F} \to \mathcal{Q} \to 0$ be a short exact sequence of coherent $\mathcal{O}_X$-modules with $\mathcal{F}$ torsion free, $\dim(\mathit{Supp}(\mathcal{Q})) = 0$, and $\dim_k H^0(X, \mathcal{Q}) \geq 2$. Then $\mathcal{F}$ is globally generated.*

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Proof. Consider the submodule $\mathcal{F}'$ generated by the global sections. By Lemma 01AN we see that $\mathcal{F}' \to \mathcal{Q}$ is surjective, in particular $\mathcal{F}' \neq 0$. Since X is a curve, we see that $\mathcal{F}' \subset \mathcal{F}$ is an inclusion of rank 1 sheaves, hence $\mathcal{Q}' = \mathcal{F}/\mathcal{F}'$ is supported in finitely many points. To get a contradiction, assume that $\mathcal{Q}'$ is nonzero. Then we see that $H^1(X, \mathcal{F}') \neq 0$. Then we get a nonzero map $\mathcal{F}' \to \omega_X$ by the universal property (Duality for Schemes, Lemma 0AWP). The image of the composition $\mathcal{F}' \to \omega_X \to \mathcal{F}$ is generated by global sections, hence is inside of $\mathcal{F}'$. Thus we get a nonzero self map $\mathcal{F}' \to \mathcal{F}'$. Since $\mathcal{F}'$ is torsion free of rank 1 on a proper curve this has to be an automorphism (details omitted). But then this implies that $\mathcal{F}'$ is contained in $\omega_X \subset \mathcal{F}$ contradicting the surjectivity of $\mathcal{F}' \to \mathcal{Q}$. $\square$